

Create Schemas

The **Mongoose Schemas** define the structure of the **MongoDB collections** used in the HouseHunt application. Each schema specifies the required **fields**, **data types**, **validation rules**, and **relationships** between different entities. These schemas ensure **data consistency**, **integrity**, and efficient interaction with the database.

UserSchema

- The **UserSchema** defines the structure for storing **user information** in the MongoDB database.
- It includes fields such as **Name**, **Email**, **Password**, **Phone Number**, **Profile Image**, and **User Role**.
- The schema supports **role-based access control**, allowing users to register as **Tenant**, **Landlord**, or **Administrator**.
- User passwords are securely stored using **bcrypt encryption**, ensuring safe authentication

```
const seedData = async () => {

  console.log('Seeding baseline users...');
  const users = await User.create([
    {
      name: 'John Tenant',
      email: 'tenant@househunt.com',
      password: 'password123',
      phone: '+91 98765 43210',
      role: 'tenant',
      profileImage: '',
    },
    {
      name: 'Jane Landlord',
      email: 'landlord@househunt.com',
      password: 'password123',
      phone: '+91 87654 32109',
      role: 'landlord',
      profileImage: '',
    },
    {
      name: 'Alex Admin',
      email: 'admin@househunt.com',
      password: 'password123',
      phone: '+91 76543 21098',
      role: 'admin',
      profileImage: '',
    },
  ]);

  const tenant = users[0];
  const landlord = users[1];
}
```

BookingSchema

- The **BookingSchema** manages **property visit bookings** made by tenants.
- It stores information such as the **Tenant**, **Property**, **Visit Date**, **Visit Time**, **Booking Status**, and **User Message**.
- The schema maintains relationships between the **User** and **Property** collections using **Mongoose references**.
- It enables efficient tracking of **pending**, **approved**, **completed**, and **cancelled** booking requests.

```
25 const seedData = > bookin Aa ab_* 4 of 6 ↑ ↓ ≡ ×
130
131 // 5. Seed a Booking Request
132 console.log('Seeding sample visit booking request...');
133 await Booking.create({
134   tenant: tenant._id,
135   property: seededProperties[1]._id, // second property
136   visitDate: new Date(Date.now() + 3 * 24 * 60 * 60 * 1000), // 3 day
137   visitTime: '11:00 AM',
138   message: 'Hello, I would love to check out this villa. I am a softw
139   status: 'pending',
140 });
141
142 console.log('\n\x1b[32m🌟 [Database Seeding Completed Successfully!]
143 process.exit(0);
144 } catch (error) {
145   console.error('\x1b[31mError seeding database:\x1b[0m', error);
146   process.exit(1);
147 }
148 };
149
150 seedData();
151
```

PropertySchema

- The **PropertySchema** stores complete information about **rental properties** listed by landlords.
- It includes details such as **Property Title, Description, Address, City, State, Price, Property Type, Bedrooms, Bathrooms, Amenities, Images, and Owner Information.**
- The schema also maintains **property status, availability, view count**, and additional features such as **parking, furnished status, and virtual tour links.**
- It enables efficient **property management, search, and filtering** within the HouseHunt application.

```
console.log('Seeding sample property listings...');
const propertiesData = sampleProperties.map((property) => {
  return {
    ...property,
    owner: landlord._id,
  };
});

// Let's add one property as "pending" for admin approval testing
propertiesData.push({
  title: "Charming Lakefront Cottage",
  description: "Escape to nature in this gorgeous wooden cottage situated directly on the lake. Perfect",
  address: "12 Whispering Pines Route",
  city: "bangalore",
  state: "karnataka",
  price: 5200,
  propertyType: "house",
  bedrooms: 2,
  bathrooms: 1.5,
  area: 1100,
  amenities: ["AC", "Parking", "Pets Allowed", "Garden", "Waterfront", "Fireplace"],
  images: [
    "https://images.unsplash.com/photo-1507089947368-19c1da9775ae?auto=format&fit=crop&w=800&q=80"
  ],
  furnished: "semi",
  parking: true,
  petsAllowed: true,
  virtualTourUrl: ""
});
```

DATABASE CODE:

```
const path = require('path');
const fs = require('fs');

const backendModulesPath = path.join(__dirname,
  '../backend/node_modules');
process.env.NODE_PATH = backendModulesPath;
require('module').Module._initPaths();
```

```
const mongoose = require('mongoose');
const dotenv = require('dotenv');

dotenv.config({ path: path.join(__dirname, '../backend/.env') });

const User = require('../backend/models/User');
const Property = require('../backend/models/Property');
const Booking = require('../backend/models/Booking');
const Review = require('../backend/models/Review');
const Wishlist = require('../backend/models/Wishlist');

const sampleProperties = JSON.parse(
  fs.readFileSync(path.join(__dirname, 'sampleProperties.json'),
    'utf-8')
);

const seedData = async () => {
  try {
    const mongoUri = process.env.MONGO_URI ||
'mongodb://127.0.0.1:27017/househunt';
    console.log(`Connecting to database: ${mongoUri}`);
    await mongoose.connect(mongoUri);
    console.log('Database connection successful!');

    console.log('Clearing old collections...');
    await User.deleteMany();
    await Property.deleteMany();
    await Booking.deleteMany();
    await Review.deleteMany();
    await Wishlist.deleteMany();
    console.log('Data cleared successfully.');
```

```
    console.log('Seeding baseline users...');
    const users = await User.create([
      {
        name: 'John Tenant',
        email: 'tenant@househunt.com',
        password: 'password123',
        phone: '+91 98765 43210',
```

```

        role: 'tenant',
        profileImage: '',
      },
      {
        name: 'Jane Landlord',
        email: 'landlord@househunt.com',
        password: 'password123',
        phone: '+91 87654 32109',
        role: 'landlord',
        profileImage: '',
      },
      {
        name: 'Alex Admin',
        email: 'admin@househunt.com',
        password: 'password123',
        phone: '+91 76543 21098',
        role: 'admin',
        profileImage: '',
      },
    ],
  ));

const tenant = users[0];
const landlord = users[1];
const admin = users[2];

console.log(`Users created:`);
console.log(`- Tenant: ${tenant.email} (password: password123)`);
console.log(`- Landlord: ${landlord.email} (password:
password123)`);
console.log(`- Admin: ${admin.email} (password: password123)`);

// 2. Initialize Empty Wishlists for users
await Wishlist.create([
  { user: tenant._id, properties: [] },
  { user: landlord._id, properties: [] },
  { user: admin._id, properties: [] },
]);

// 3. Create Sample Properties owned by Jane Landlord
console.log('Seeding sample property listings...');
const propertiesData = sampleProperties.map((property) => {
  return {
    ...property,

```

```

        owner: landlord._id,
      };
    });

    // Let's add one property as "pending" for admin approval testing
    propertiesData.push({
      title: "Charming Lakefront Cottage",
      description: "Escape to nature in this gorgeous wooden cottage
situated directly on the lake. Perfect weekend getaway with a large
boat dock, wood-burning stove, and glass-paneled sunroom.",
      address: "12 Whispering Pines Route",
      city: "bangalore",
      state: "karnataka",
      price: 5200,
      propertyType: "house",
      bedrooms: 2,
      bathrooms: 1.5,
      area: 1100,
      amenities: ["AC", "Parking", "Pets Allowed", "Garden",
"Waterfront", "Fireplace"],
      images: [
        "https://images.unsplash.com/photo-1507089947368-19c1da9775ae?auto=form
at&fit=crop&w=800&q=80"
      ],
      furnished: "semi",
      parking: true,
      petsAllowed: true,
      virtualTourUrl: "",
      status: "pending", // Pending approval!
      owner: landlord._id,
      views: 12
    });

    const seededProperties = await Property.create(propertiesData);
    console.log(`${seededProperties.length} property listings
seeded.`);

    // 4. Seed a Review
    console.log('Seeding sample review...');
    await Review.create({
      user: tenant._id,
      property: seededProperties[0]._id, // first property

```

```

        rating: 5,
        comment: "Absolutely breathtaking penthouse! The view is
incredible, the marble finishes are exquisite, and Jane was a fantastic
landlord. Highly recommended!",
    });

    // 5. Seed a Booking Request
    console.log('Seeding sample visit booking request...');
    await Booking.create({
        tenant: tenant._id,
        property: seededProperties[1]._id, // second property
        visitDate: new Date(Date.now() + 3 * 24 * 60 * 60 * 1000), // 3
days in future
        visitTime: '11:00 AM',
        message: 'Hello, I would love to check out this villa. I am a
software engineer and looking for a quiet workspace.',
        status: 'pending',
    });

    console.log('\n\x1b[32m★ [Database Seeding Completed
Successfully] ★\x1b[0m\n');
    process.exit(0);
} catch (error) {
    console.error('\x1b[31mError seeding database:\x1b[0m', error);
    process.exit(1);
}
};

seedData();

```